

Tactical Display Mockups

Six LCD apps for the Rotating Antenna, drawn from one shared set of seven made-up contacts. The radar, the list and the track screen show the same ship because the real apps will read the same contact table.

All values are illustrative. A dash on any screen means the game cannot supply that field. It never means zero.

The scenario

Our ship is cruising low over a planet at 64 m/s with a dish on the mast. This table is the data behind every panel below. The last column is the rule from the detection plan that put each contact on the screen.

ID	SHOWN AS	DETECTED BY	CLASS	RANGE	BRG	ELEV	SPEED	CLOSING	REL	STATE
...3392	UNIDENTIFIED	rule 3: inside 2 km, not broadcasting	SMALL SHIP	1.40 km	312°	+1.2°	9 m/s	+34 m/s	–	DETECTED
...6130	UNIDENTIFIED	rule 3: inside 2 km, not broadcasting	LARGE SHIP	1.90 km	118°	+21.0°	1 m/s	–29 m/s	–	DETECTED
...2284	TUG AURORA	rule 2: owned by us	SMALL SHIP	3.12 km	255°	+1.1°	26 m/s	–1 m/s	OWN	DETECTED
...5207	VANGUARD-3	rule 1: antenna, 15 km radius	LARGE SHIP	4.21 km	037°	+5.7°	55 m/s	+68 m/s	HOSTILE	TRACKED
...8845	HAULER K	rule 1: antenna, 15 km radius	LARGE SHIP	5.50 km	205°	–2.0°	34 m/s	–	NEUTRAL	LOST 11 s
...9051	SALVAGE 7	rule 1: beacon, 20 km radius	SMALL SHIP	6.38 km	084°	–8.1°	6 m/s	+6 m/s	NEUTRAL	DETECTED
...0716	OUTPOST DELTA	rule 1: antenna, 50 km radius	STATION	11.80 km	160°	–1.3°	0 m/s	–60 m/s	FRIENDLY	DETECTED

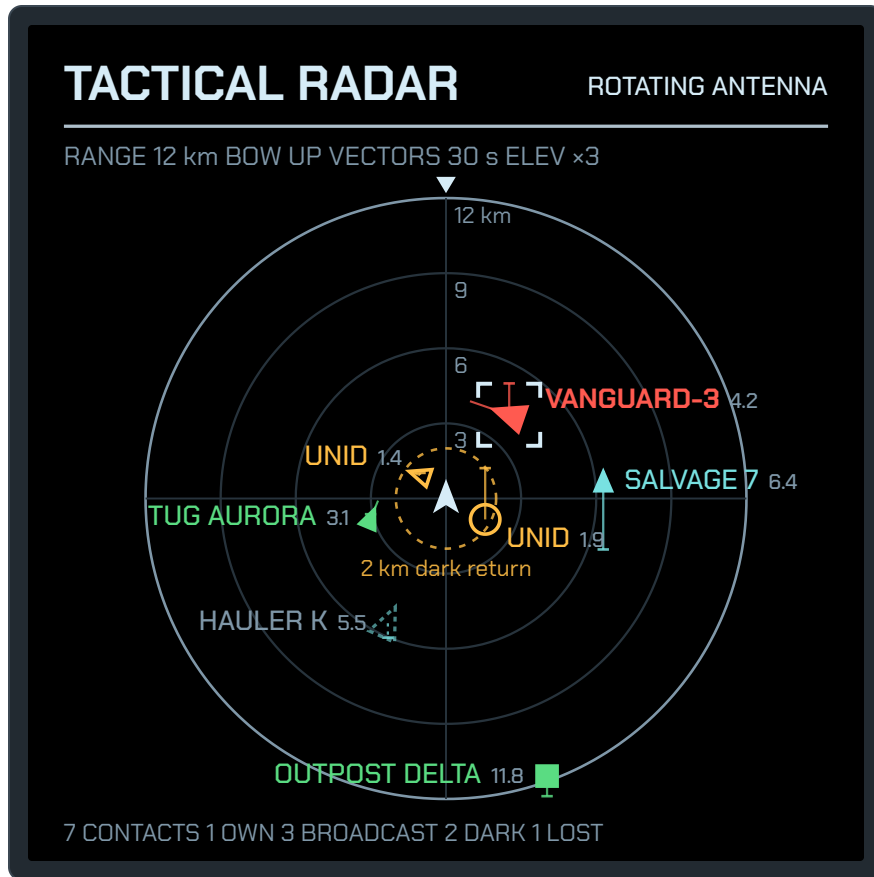
Reading the markers

-  Square: a station. It does not move.
-  Circle: a ship that is not really moving, so it has no course.
-  Brackets: the tracked contact.
-  Large triangle: a large ship, pointing along its course.
-  Hollow amber: a dark return. Seen, not identified.
-  Dashed ring: the 2 km dark return boundary.
-  Small triangle: a small ship.
-  Dashed, dim: lost. Drawn at its last known position.
-  Colour: green own or friendly, cyan neutral, red hostile, amber unknown.

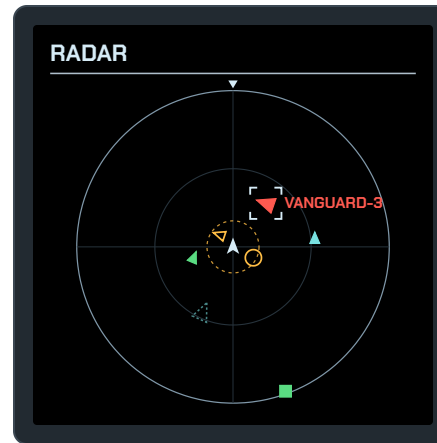
Tactical Radar

`GT_Radar` reads range, bearing, elevation, velocity, class and state for every contact, plus the tracked one.

- Bow is up. Each contact sits at its horizontal range. The stalk shows how far it is above or below the ship, at three times scale.
- The dashed amber ring is 2 km. Inside it the dish sees everything. Outside it, only broadcasters and your own ships appear.
- Filled markers are identified. Hollow amber markers are dark returns. Dashed markers are lost contacts, drawn where they were last seen.
- On a small LCD the same app drops labels, stalks and vectors, and keeps the rings, the markers and the tracked contact.



512 × 512 surface. VANGUARD-3 is the tracked contact, shown with brackets.



Small LCD. Same app, less detail.

Contact List

GT_Contacts reads the same contact table as the radar, one row each.

- Rows run in range order. Sort, columns and row count are set in the panel's Custom Data.
- Dark returns have a size class and a speed but no name and no relationship. They never show a guess.
- A lost contact stays in the list, dimmed, until its retention time runs out.
- When the surface is too short for every row, the last line becomes "+N more".

CONTACTS

ROTATING ANTENNA

SORT RANGE ASC 7 OF 7 SHOWN

ID	NAME	RNG	BRG	CLS	SPD	REL
▼...3392	UNIDENTIFIED	1.40	312	SML	9	—
○...6130	UNIDENTIFIED	1.90	118	LRG	1	—
◀...2284	TUG AURORA	3.12	255	SML	26	OWN
▶...5207	VANGUARD-3	4.21	037	LRG	55	HOS
⚡...8845	HAULER K	5.50	205	LRG	34	LOST
▲...9051	SALVAGE 7	6.38	084	SML	6	NEU
■...0716	OUTPOST DELTA	11.80	160	STN	0	FRN

RNG km BRG degrees from the bow SPD m/s

REL own, friendly, neutral, hostile

A dash means not available. Lost rows are last known.

1 LOST HAULER K expires in 19 s

512 × 512 surface. The tracked row is highlighted.

Target Track

GT_Track reads the tracked contact in full.

- The big numbers are range and closing rate. The dial and the elevation scale show where it is relative to the bow.
- A field the game cannot supply reads as a dash. Here the broadcaster does not share a ship name, so that row is empty.
- No threat assessment appears anywhere. The sensor reports position, motion, size and what the target broadcasts.
- When the contact drops off, the screen keeps its last-known values dimmed and counts down the retention time.



Tracked. Every field is live.

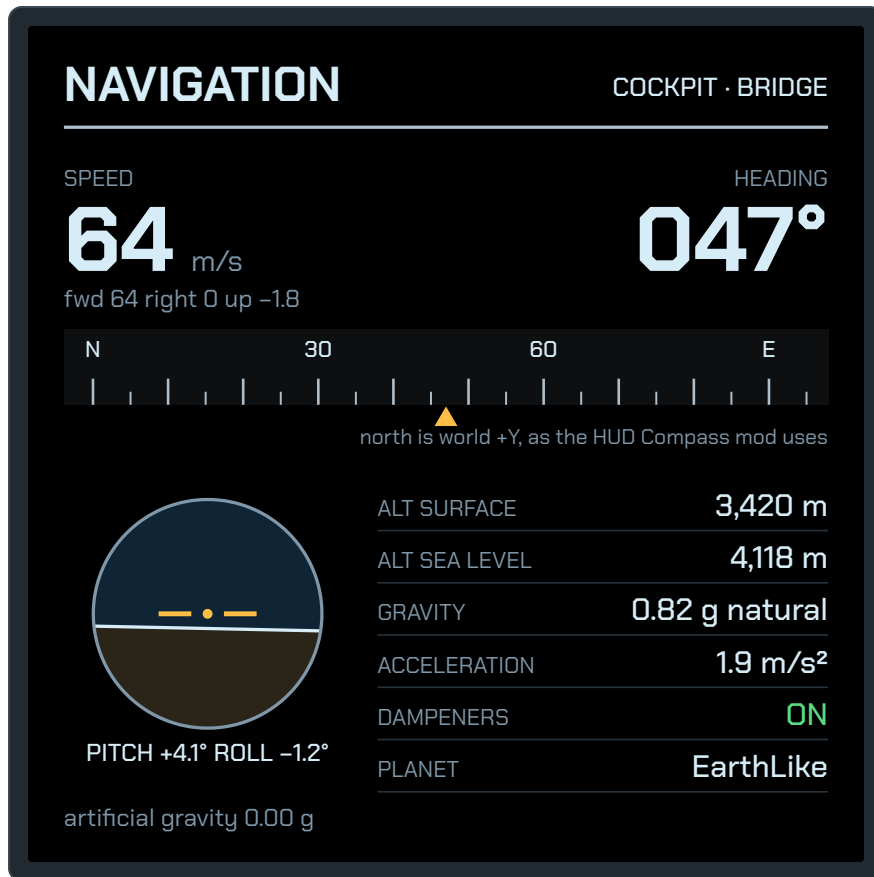


Lost. Last-known values, closing rate unavailable.

Navigation

GT_Navigation reads a cockpit, remote control or flight seat on the same grid.

- Speed, heading, altitude, gravity and attitude come from the ship controller, not from the dish.
- Heading uses the same north as the HUD Compass mod, so it agrees with what players already see.
- With no controller on the grid the app says so and shows dashes. It does not borrow a heading from the antenna.



Flying low over a planet.



No controller. Nothing is estimated.

Sensor Status

`GT_SensorStatus` reads the state the tactical service actually holds.

- Nothing here is a message written for effect. Each line is a number or a flag the service keeps.
- The streaming limit is the game's sync distance. The screen says when it is the limit that is hiding contacts.
- The 2 km dark return range is a fixed rule, not a setting, so it reads the same on every ship.

SENSOR STATUS

ROTATING ANTENNA

ONLINE

2 of 2 dishes working

DARK RETURN RANGE

fixed

2.0 km

STREAMING LIMIT

not binding

15.0 km

SCAN INTERVAL

2.0 s

LAST SCAN

0.6 s ago

CONTACTS

1 own 3 broadcast 2 dark 1 lost

7 / 64

MIN GRID SIZE

smaller grids are ignored

5 blocks

ERROR

none

Online. Seven contacts, sync distance not limiting.

SENSOR STATUS

ROTATING ANTENNA

DEGRADED

1 of 2 dishes working

DARK RETURN RANGE

fixed

2.0 km

STREAMING LIMIT

binding: nothing beyond it is detected

3.0 km

SCAN INTERVAL

2.0 s

LAST SCAN

one dish has no power

7.4 s ago

CONTACTS

1 own 1 dark

2 / 64

MIN GRID SIZE

smaller grids are ignored

5 blocks

ERROR

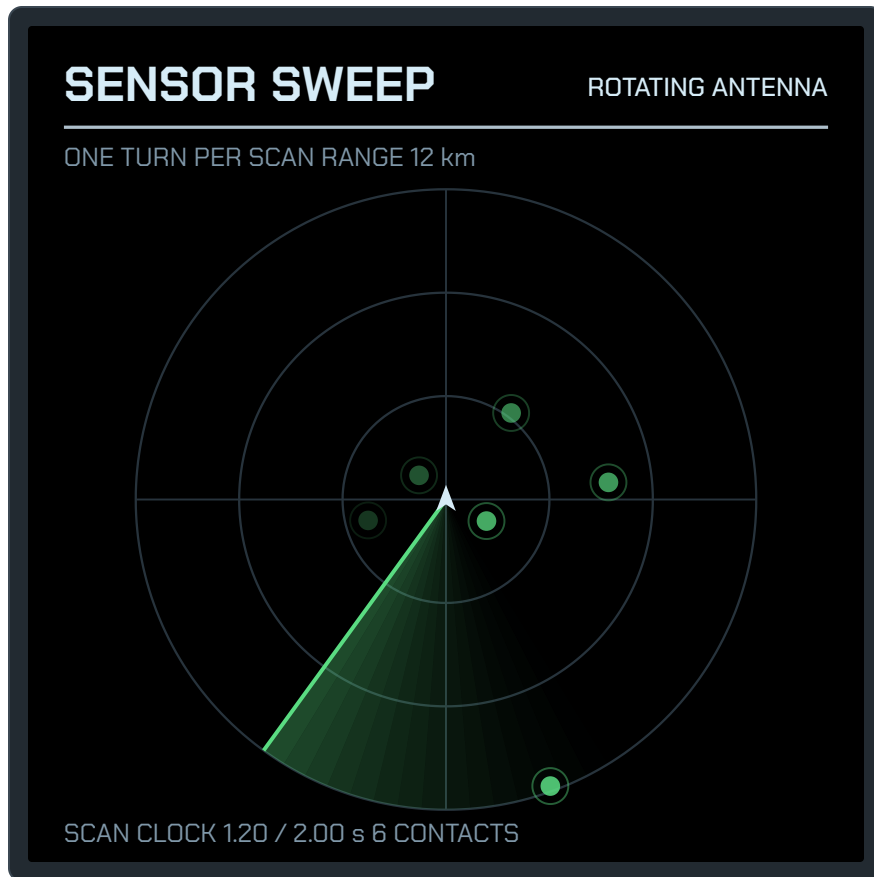
Aft Mast: no power

Degraded. A dish has lost power and the sync distance is short.

Sensor Sweep

`GT_Sweep` reads the scan clock and the same contact table.

- The sweep turns once per scan interval, so its angle is the time since the last scan. It is a clock, not a beam.
- It does not aim the dish and nothing is detected because it passes. The dish keeps spinning as it does today.
- Contacts brighten as the sweep passes and fade toward the next scan. Lost contacts are not drawn, because nothing is observed.
- Throttled on its own and never run on a dedicated server.



1.2 s into a 2.0 s scan. HAULER K is not drawn because it is lost.

Questions these mockups raise

1. Dark returns all read UNIDENTIFIED. With three of them in range the only difference is the ID tail and the range. A local track number (U1, U2) would help, but the game does not provide one, and it would reset on reload. Worth adding?
2. Elevation stalks are drawn at three times scale and say so in the header. At 12 km, 700 m of height is 11 pixels without that. Is the printed multiplier clear enough, or should stalks only appear on the tracked contact?
3. On the radar, colour is the only cue for hostile or friendly. The list spells it out in letters. Putting a letter beside each plotted marker would cover colour-blind players, at the cost of clutter.
4. The list shows relative bearing only. A true bearing column needs a compass convention on screen, and the Navigation app already has to pick one.